

RoomReady: Study Room Reservation System - Vision Document v1.0

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for

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Problem Statement

Many library patrons struggle to find a dedicated study space due to high demand and complex booking rules. At Round Rock Public Library, visitors can reserve rooms for 1 hour daily, twice weekly (City of Round Rock, n.d.). With only two youth rooms and six adult rooms requiring reservations a week in advance, patrons often face disappointment when rooms are unavailable or when they have exceeded their weekly limit. This wastes valuable time for study and collaboration (Stonebraker, 2016; Wei Xuan, 2021). Library staff also manage these spaces, handling conflicts, verifying resident eligibility, tracking borrowed equipment like HDMI cables, enforcing noise and capacity limits (four per room), and locating lost items (City of Round Rock, n.d.).

Target Users

User Type	Description	Primary Goal	Technical Level	Usage Frequency
Library Patrons	Public library users who need a quiet room for studying, tutoring, remote work, or small meetings.	Quickly see which study rooms are available and reserve one.	Low-Moderate	Daily
Library Staff	Staff who assist patrons and monitor room usage during library hours.	View current room status and help patrons reserve open rooms.	Low-Moderate	Daily
Library Event Coordinator	The library manager or coordinator is responsible for understanding room demand and space usage.	Monitor room status and identify patterns in study room use	Moderate	Weekly/Monthly

Table 1. RoomReady Target User Profiles: Goals, Technical Proficiency, and Interaction

Frequency.

Proposed Solution

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RoomReady is a kiosk-style web application that helps library patrons quickly identify and reserve available study rooms. The proof of concept demonstrates a simple end-to-end flow: a user views room availability, selects an available room, submits a reservation form, and sees the room status update from Available to Reserved (Sommerville, 2016). The system does not attempt to manage future date/time reservations, recurring bookings, patron authentication, check-ins, or library-wide event scheduling in this version. By focusing the design on an intuitive, touch-friendly kiosk user interface, the platform directly minimizes patron navigation friction and reduces cognitive load during immediate space procurement (Perera et al., 2024).

Key Features

#	Key Feature	Description
1	Study Room Availability Dashboard	Displays public library study rooms with a clear Available or Reserved status.
2	Room Selection	Allows patrons to select a room marked Available.
3	Self-Service Cancellation	One-click cancellation links to quickly free up in-demand rooms.
4	Reservation Form	Collects basic reservation details for the selected study room.
5	Reservation Confirmation	Displays a clear confirmation message so the patron knows the reservation was completed.

Table 2. *RoomReady System Capabilities and Functionality Overview: Core Features of the RoomReady Platform.*

Success Criteria

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#	Success Criterion	How it Will Be Verified
1	Core Flow Efficiency: A new patron can view availability and complete a valid room reservation in under 3 minutes via the kiosk interface (Perera et al., 2024).	Verified via timed usability trials during the live demonstration walkthrough and local UI interaction benchmarking.
2	Rule & Quota Enforcement: The system successfully blocks 100% of reservation attempts that violate the daily time or weekly frequency limits (City of Round Rock, n.d.).	Verified through automated boundary-value test suites, backend endpoint assertions, and mock API data injections.
3	Self-Service Autonomy: A user can successfully cancel their own reservation without staff intervention 100% of the time.	Demonstrated via end-to-end testing of the automated cancellation state routes and token/link validation checks.
4	Operational Efficiency: Library staff spend 50% less time mediating walk-up booking conflicts and explaining reservation policies to patrons.	Evaluated via simulated operational time-studies and baseline comparisons against manual logging workflows.
5	Public Accessibility: The application is deployed to a public URL and is fully accessible 24/7.	Live production URL submitted on Blackboard with uptime monitoring enabled.
6	Data Synchronization: The frontend accurately and consistently reflects real-time changes in room status (Available vs. Reserved) upon form submission.	Verified via Postman API testing, route integration checks, and local state assertion.
7	Technical Reproducibility: The project repository structure enables a seamless, standalone deployment from a clean environment.	Verified by a peer or instructor attempting to set up the app cold using instructions in /poc/README.md.

8	Audit Traceability: 100% of software developments, architectural decisions, and requirement additions are documented cleanly.	Visible in the GitHub commit history using Conventional Commits and PR peer reviews.
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Table 3. *Measurable Performance Benchmarks and Quality Assurance Metrics*

Constraints

Category	Description
Constraint	Timeline: The Proof of Concept (POC) must be designed, built, and submitted within the 5-hour Saturday session.
Constraint	Team Size: The project is strictly limited to the 3-person engineering team (Product Owner, Architect, Dev Lead).
Constraint	Technology Scope: The POC must demonstrate a single, end-to-end user flow, either as a frontend prototype or a mini full-stack app.
Assumption	Technical Environment: The POC will run in a standard web browser and requires basic internet connectivity to access the deployment.
Assumption	Tool Proficiency: Team members will leverage AI coding assistants to accelerate development while maintaining a full understanding of all generated code.
Constraint	Version Control: All code integration requires adherence to Git branch protection rules and ensures all changes are documented via Conventional Commits.

Table 4. *RoomReady Project Guardrails: Structural Constraints and Environmental*

Assumptions.

Out of Scope

To keep the proof of concept feasible and reliable within the sprint timeframe, the following items are excluded from the current version:

- *Future Date and Time Reservations:* RoomReady will show only current availability. It will not include calendar scheduling or one-hour time blocks in the proof of concept.

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- *Check-In Workflow:* The system will not require patrons to check in after reserving a room.
- *Authentication:* Patrons will not need to create accounts or log in for the prototype.
- *Recurring Reservations:* The system will not support repeating reservations.
- *Room Capacity Matching:* The prototype will not filter rooms by party size, group size, or seating capacity.
- *Payment Processing:* The system will not handle payments or fees.
- *Native Mobile Application:* RoomReady will be built as a responsive web/kiosk prototype, not a native iOS or Android app.
- *Full Production Backend:* The prototype will use mock data or local frontend state rather than a production database.

Ethical Use of AI

The team used AI tools to support brainstorming, document structure, frontend scaffolding, and troubleshooting during the sprint. All final project decisions, scope choices, code commits, and presentation explanations were reviewed and owned by the team. The team is prepared to explain the purpose and behavior of all code included in the RoomReady proof of concept.

References

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VISION DOCUMENT

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